Baryogenesis in the Psion Hypothesis: Psion-Number Conservation, No-Go Results, and Level Bifurcation at Reheating as a Candidate Mechanism

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Abstract

The Psion Hypothesis (PH) builds all matter from a single six-label fermion, the psion, interacting through one four-fermion Lagrangian. We examine what this structure implies for the baryon asymmetry of the Universe. Four results are exact. (i) Psion number is conserved by every term of the Lagrangian, and the continuous internal symmetry is only $U(1)_N \times U(1)_M$. (ii) Field-theory CP is exact. (iii) Observed reactions select a unique ontology for antimatter, composites of antipsions, and require exactly two additive charges, B and L, with $N_\psi/3 = B+L$; the PH therefore conserves $B+L$ and allows only $B-L$ to change, the reverse of the Standard Model. (iv) The Lagrangian supplies B neither as a Noether charge nor as a topological charge, and the cascade level is not an internal degree of freedom. On this basis we show, with pre-registered numerical tests, that neither the Lagrangian nor its minimal extension produces the asymmetry. We then formulate a candidate mechanism, *level bifurcation*: each psion triple formed after inflationary reheating binds into a baryonic or a leptonic rung with a CP-odd bias ε, and the rung is conserved afterwards. If the mechanism holds, $B = -L$, charge neutrality and a final state of hydrogen plus two net antineutrinos per baryon follow from its symmetry alone. With the natural coupling of the Lagrangian, the departure from equilibrium is set by the reheating temperature. The efficiency peaks at $\eta_B/\varepsilon \simeq 3.6\times10^{-3}$ for $T_{\rm RH} \simeq 1.4\times10^{10}$ GeV, and $\varepsilon \leq 10^{-5}$ suffices for $T_{\rm RH} \approx 2\times10^{9}$–$2\times10^{10}$ GeV. This window lies below the condensation scale $\Lambda_{\rm meta} \simeq 1.3\times10^{12}$ GeV, so a single orientation of the condensate can fix the sign of the bias over the observable Universe. The mechanism rests on two conditions still to be derived: a conserved rung label and the magnitude of the bias. The PH also makes structural predictions independent of the mechanism: no nucleon decay with $\Delta B = \Delta L$, no $0\nu\beta\beta$ and no n–$\bar{n}$ oscillation.

1 Introduction

A dynamical origin of the baryon asymmetry, $\eta_B = n_B/n_\gamma = 6.1\times10^{-10}$ [1], requires baryon-number violation, C and CP violation, and departure from thermal equilibrium [2]. Successful frameworks, such as electroweak baryogenesis [3], leptogenesis [4, 5] and the Affleck–Dine mechanism [6], derive all three ingredients within a specified model.

The Psion Hypothesis [7] is a single-Lagrangian framework in which the observed fermions are three-psion composites on a mass cascade $m(\ell) = \Lambda_{\rm meta} e^{-\ell/b_e}$. Earlier versions assigned the asymmetry to a chirality parameter $\varepsilon_\psi = 25/169$ combined with a Standard-Model sphaleron conversion. Here we take the Lagrangian at face value and ask what it permits. The answer is restrictive. The constraints turn out to be strong enough to fix the *form* a viable mechanism must take, and we formulate and test one such mechanism.

The paper proceeds from exact statements to conditional ones. Section 2 recalls the framework. Section 3 proves the conservation structure. Section 4 closes the mechanisms available within the Lagrangian and its minimal extension. Section 5 formulates level bifurcation and states the conditions under which it accounts for η_B. Section 6 collects the predictions. Every numerical test was pre-registered: the definition, with a SHA-256 hash and timestamp, was fixed before execution, and all outputs, including failed controls, are public [8].

2 Framework

The Lagrangian is

$$\mathcal{L} = \bar{\psi}\, i\gamma^\mu \partial_\mu \psi - G_{\rm UV}(\bar{\psi}\psi)^2 - (\bar{\psi} J^a \psi)(\bar{\psi}\Sigma^a \psi), \tag{1}$$

where ψ carries a six-valued label $k = 0, \ldots, 5$ (internal spin $j = 5/2$, $m_k = k - 5/2$). J^a and Σ^a are the $\mathrm{SU}(2)_{\rm meta}$ generators restricted to the left ($k \le 2$) and right ($k \ge 3$) sectors. The labels carry charges $Q_k = (1, -5, -5, 10, 4, 4)/9$. Composites of three distinct labels ("triples") number 20. Among them are the electron $(0, 1, 2)$, the proton $(0, 4, 5)$ and the neutron $(0, 1, 5)$. The cascade level ℓ is an independent quantum number: the nucleon scale is $\ell = 8$ (610 MeV) and the electron $\ell = 10$ [7]. Three neutral triples are listed in [7] as dual-sector, each being either a baryon or a neutrino: $(0, 1, 5) = n/\nu_\tau$, $(0, 2, 4) = \Sigma^0/\nu_\mu$, $(1, 2, 3) = \Lambda^0/\nu_e$. The condensation scale is $\Lambda_{\rm meta} = 1.26 \times 10^{12}$ GeV.

3 Exact conservation structure

Theorem 1 (Psion number)**.** *N_ψ, the charge of $\psi \to e^{i\alpha}\psi$, is conserved by every term of* (1)*. In the condensed phase it is the only exact additive charge.*

The first statement is immediate. The second follows from an exact charge audit over the 20 triples and four condensate variants, which leaves no further candidate [8].

Theorem 2 (Continuous internal symmetry)**.** *The Lie algebra of transformations $\psi \to e^{tX}\psi$ that leave the interaction in* (1) *invariant is two-dimensional and abelian. It is generated by the identity and by* $\mathrm{diag}(m_k)$*, i.e. the group is $U(1)_N \times U(1)_M$ with $M = \sum m_k$. No continuous chiral extension ($X \otimes \gamma_5$) survives the $G_{\rm UV}$ term. The winding condensate $J^+\Sigma^+ + J^-\Sigma^-$ breaks $U(1)_M$ and preserves $U(1)_N$.*

The invariance condition is linear in X. Its solution space was computed exactly in the 36-dimensional space of anti-Hermitian 6×6 matrices.

Proposition 3 (CP)**.** *The coupling tensors of* (1) *are real and Hermitian, so field-theory CP is exact. The chirality parameter ε_ψ is even in any hopping phase, hence CP-even. The six Z6 vacua violate CP pairwise, but conjugate vacua $(k, 6-k)$ are degenerate.*

Theorem 4 (Antimatter and the observed charges)**.** *Consider the two readings of antimatter: (O^+) a triple of psions with inverted labels $k \to (6-k) \bmod 6$; (O^-) the same triple of antipsions, with $N_\psi = -3$ and charge $-Q$. Test both against the observed reactions $n \to p e^- \bar{\nu}_e$, $p \to n e^+ \nu_e$, $e^- p \to n \nu_e$, $e^+ e^- \to 2\gamma, 3\gamma$, $\gamma \to e^+ e^-$, $p\bar{p} \to \pi^+\pi^-\pi^0$, $\pi^0 \to 2\gamma$, $\pi^+ \to \mu^+\nu_\mu$ and $\mu^- \to e^-\bar{\nu}_e\nu_\mu$. Reading O^+ conserves neither N_ψ nor charge and assigns $\bar{\Sigma}^+$ the charge $+1$. Reading O^- passes all tests, with $N_\psi = 0$ for every boson.*

Theorem 5 (Charges required by observation)**.** *Treat the cascade class as a label: H (level 8: p, n, Λ, Σ^0), E (level 10 and its excitations: e, μ), ν (three neutrinos). The additive charges conserved by the observed reactions $\beta^{\pm}$ decay, e^- capture, $\pi^+ \to \mu^+\nu_\mu$, μ decay, $\nu_\mu n \to \mu^- p$, $\bar{\nu}_e p \to e^+ n$, $\Lambda \to p\pi^-, n\pi^0$, $\Sigma^0 \to \Lambda\gamma$ and neutrino oscillations span exactly a two-dimensional space. Its basis is B (number of H states) and L (number of E and ν states), and $N_\psi/3 = B + L$.*

Corollary 6. *Since N_ψ is exact, $B + L$ is conserved. Baryogenesis in the PH requires a change in $B - L$, the reverse of the Standard Model, where $B - L$ is conserved and sphalerons change $B + L$. The Standard-Model sphaleron conversion therefore does not apply.*

Theorem 7 (Baryon number is not available in (1))**.** *(a) B is not in the span of the Noether charges (M, Q, N_ψ) evaluated on the observed states. (b) The vacuum manifold of the condensed phase has $\pi_2 = \pi_3 = 0$, so no point-like topological charge exists and there is no analogue of the Skyrme baryon number [9]. (c) Composite states are labelled by their triple alone; the cascade level carries no internal quantum number.*

Corollary 8 (Dual-sector instability)**.** *With only N_ψ and Q conserved, the processes $p \to \nu\pi^+$, $n \to \nu\pi^0$, $n \to 3\nu$, $pe^- \to \nu_\mu\nu_\tau$ and the same-triple level jumps $n \to \nu_\tau\gamma$, $\Lambda^0 \to \nu_e\gamma$, $\Sigma^0 \to \nu_\mu\gamma$ are all allowed, and all have $\Delta B = -\Delta L$. The measured limits are $p \to \bar{\nu}\pi^+ > 3.9 \times 10^{32}$ yr and $n \to \bar{\nu}\pi^0 > 1.1 \times 10^{33}$ yr [10], and $n \to$ invisible $> 5.8 \times 10^{29}$ yr [11]. They require an amplitude suppression $\lesssim 10^{-8}$ relative to a contact operator at $\Lambda_{\rm meta}$. Same-triple jumps, which no label-changing selection rule of (1) can forbid, require $\lesssim 5 \times 10^{-30}$ relative to an electromagnetic transition (reference lifetime 10^{29} yr). A conserved baryon label must therefore be added to the PH.*

The count of neutral triples is five, not three: $(0, 1, 4)$ and $(0, 2, 5)$ are also neutral. They carry the residual discrete charge $D = (2M - 6Q + 3) \bmod 4 = 2$, which is exact in the condensed phase.

4 No-go results

Emergent sphalerons. Any sphaleron-like configuration of the PH is built from psions and conserves N_ψ, hence $B + L$. It cannot convert a lepton asymmetry into baryons in the Standard-Model manner.

Temperature histories. CPT gives particle and antiparticle equal masses and equal equilibrium abundances. With exact CP and a symmetric initial state, a Boltzmann network of the cascade levels returns zero asymmetry for any programme of cooling and reheating; reheating only dilutes an existing asymmetry. Temperature controls the composition across levels, not the matter–antimatter ratio.

Z6 domains with spontaneous CP. Inside a CP-violating domain, assembly of composites out of equilibrium does generate an asymmetry, $\Delta p \propto \varepsilon$, together with an equal antipsion reservoir. However, the Z6 walls form at $T \approx 3$ MeV, when the assembly rate is $\Gamma/H \sim 10^{-37}$. A domain's maximal comoving size today is ~ 30 pc, against the $\gtrsim 1$ Gpc required of antimatter regions [12]. And degenerate conjugate vacua give a zero global average.

CP terms built from PH operators. We tested 16 CP-odd patterns built from the PH's own operators (commutators of Q, M, n_R, the Z6 index and the energy with h and $J \cdot \Sigma$, and products in rearrangements), singly and in pairs. Fitted to the observed composition, none performs better than random directions. Inserting a phase directly, $J \cdot \Sigma \to \mathrm{Re}(e^{-i\varphi} O)$, with full density-matrix

assembly kinetics, generates an asymmetry with $\varphi \sim 10^{-4}$–10^{-2}. But exotic triples receive 2–2500 times the proton asymmetry, and the electron asymmetry has the wrong sign in most runs. Without a conserved label, no CP source selects the observed composites.

Minimal extension. A two-valued species index $\psi \to \psi_s$, $s \in \{B, L\}$, with currents summed over species and $SU(2)_{\text{species}}$ broken, makes B and L Noether charges. It allows all observed reactions and forbids the processes of Corollary 8 exactly. It preserves the PH's numerical outputs only if the index does not enter the one-loop coefficient $|\beta_0| = N_{\text{Dirac}} + 2/N_f$: doubling N_f gives $m_e = 0.065$ MeV instead of 0.509 MeV. Because the label is carried by the constituents, triples cannot choose their sector. $B - L$ can then change only through species mixing, and the proton lifetime bounds the mixing coefficient to $c \leq 1.3 \times 10^{-8}$. Mixing is then active only above 2.7×10^{15} GeV, outside the range of the contact description.

Proposition 9. *Neither* (1) *nor its minimal species extension produces the baryon asymmetry. A viable mechanism needs a CP-odd ingredient acting on a conserved label that the constituents do not carry.*

5 Level bifurcation at reheating

5.1 Postulates

The dual sector, which causes the instability of Corollary 8, also supplies the structure that the failed mechanisms lacked: two final channels from a common initial state. That is the minimal setting in which CP violation is compatible with CPT and unitarity [13]. We formulate the mechanism as three postulates.

Postulate 1 (Rung label)**.** *A psion triple binds into the baryonic rung (t_B) or the leptonic rung (t_L). The rung is acquired at binding, not carried by the constituents. It is conserved by every process except dissociation of the triple into free psions.*

Postulate 2 (Biased bifurcation)**.** *The two channels are populated with a CP-odd bias ε, of opposite sign for antipsion triples. The sign is fixed by the winding orientation of the condensate. The CP source is ε times the net formation rate, so it vanishes in equilibrium; dissociation followed by re-formation washes out the channel imbalance.*

Postulate 3 (Formation after reheating)**.** *Triples form in the thermal bath after inflationary reheating at $T_{\text{RH}} < \Lambda_{\text{meta}}$, with the natural contact coupling of* (1)*.*

Postulate 1 is the conserved label that nucleon stability requires independently (Corollary 8). Postulate 3 has a second consequence. Because $T_{\text{RH}} < \Lambda_{\text{meta}}$, the condensate is not restored after inflation, and the orientation selected before or during inflation is common to the entire observable Universe. The global sign required by Postulate 2 is then a property of the vacuum rather than of local domains, which avoids the failure of Z6 domains in Section 4.

5.2 Kinetics

For each triple t and formation rate k_t, the matter equations read

$$\frac{dt_B}{dx} \ni \tfrac{1}{2} k_t P_t - \frac{k_t}{K} t_B + \tfrac{1}{2} \varepsilon J_t, \qquad \frac{dt_L}{dx} \ni \tfrac{1}{2} k_t P_t - \frac{k_t}{K} t_L - \tfrac{1}{2} \varepsilon J_t, \tag{2}$$

with $x = \ln(T_{\rm RH}/T)$, P_t the product of the three psion abundances and $J_t = k_t[P_t - (t_B + t_L)/K]$ the net formation rate. The antimatter equations are the same with $\varepsilon \to -\varepsilon$. The network also contains transitions within each rung, rearrangements $ab \to cd$ within each rung, annihilation $t_r\bar{t}_r \to$ radiation, and $\psi\bar{\psi}$ annihilation; the rates and binding weights are those of the assembly network of the PH audit [8]. All rates scale as $\Gamma/H = c_\Gamma\, G_4 T^3 M_{\rm Pl}/(1.66\sqrt{g_*}\,\Lambda_{\rm meta}^4)$, with $c_\Gamma = 1$ for the natural coupling.

Proposition 10 (Structural consequences)**.** *For a symmetric initial state, Postulates 1–2 imply, at all times,*

$$B = -L, \qquad Q_{\rm tot} = 0, \qquad \Delta N_\psi = 0. \tag{3}$$

If p, e^- and the neutrinos are the lightest states of their sectors, the late-time state is hydrogen plus two net antineutrinos per baryon, whatever the composition at formation.

Proof. The map $(\psi, t_B, t_L, \bar{t}_B, \bar{t}_L) \to (\bar{\psi}, \bar{t}_L, \bar{t}_B, t_L, t_B)$ leaves the equations invariant: it exchanges $+\varepsilon$ with $-\varepsilon$ and pairs $t_B\bar{t}_B$ with $\bar{t}_L t_L$ in the annihilation terms. It also leaves the symmetric initial state invariant. Hence $t_B = \bar{t}_L$, $t_L = \bar{t}_B$ and $\psi = \bar{\psi}$ at all times, which gives $B = \sum(t_B - \bar{t}_B) = -\sum(t_L - \bar{t}_L) = -L$ and $\Delta N_\psi = 0$. The charge of each triple enters B and L with opposite signs, so $Q_{\rm tot} = 0$. With B, L and Q conserved, baryonic excesses decay to protons and leptonic ones to e^- and neutrinos. Neutrality gives $N_e = N_p = B$, and $L = N_e - (N_{\bar{\nu}} - N_\nu) = -B$ gives $N_{\bar{\nu}} - N_\nu = 2B$. □

Proposition 10 removes the composition problem of Section 4. Exotic triples need not be absent at formation; they decay into the observed states because the labels are conserved. We verified the proposition numerically in a 92-species network. In all runs with $|B|$ above numerical noise, $|B + L|/|B| \lesssim 10^{-8}$, $|Q_{\rm tot}| \lesssim 10^{-14}$ and $|\Delta N_\psi| \lesssim 10^{-13}$. A bias of $\varepsilon = 0.1$ produces no asymmetry from an equilibrium initial state ($\lesssim 10^{-16}$), and the response is linear in ε to 3×10^{-8}.

5.3 Efficiency and the reheating temperature

With $c_\Gamma = 1$, formation reaches equilibrium ($\Gamma/H = 1$) at $T_{\rm eq} = 4.3 \times 10^9$ GeV. The departure from equilibrium is therefore set by $T_{\rm RH}$ alone, with no free rate parameter: a rate normalisation c_Γ at a given $T_{\rm RH}$ is equivalent to $c_\Gamma = 1$ at $c_\Gamma^{1/3} T_{\rm RH}$ (verified to 7×10^{-8}). Table 1 gives the efficiency.

$T_{\rm RH}$ [GeV]	7×10^{10}	4×10^{10}	2×10^{10}	1.4×10^{10}	4×10^9	1.4×10^9
Γ/H at $T_{\rm RH}$	4.2×10^3	7.8×10^2	98	32	0.78	0.033
η_B/ε (L1)	$< 10^{-12}$	1.4×10^{-6}	3.3×10^{-3}	3.6×10^{-3}	4.5×10^{-4}	2.1×10^{-5}
η_B/ε (L2)	$< 10^{-13}$	4.4×10^{-6}	2.6×10^{-3}	2.6×10^{-3}	4.3×10^{-4}	2.1×10^{-5}

Table 1: Efficiency of level bifurcation with the natural coupling, as a function of the reheating temperature, for the two binding laws of the network. Above $\sim 4 \times 10^{10}$ GeV formation is in equilibrium and the asymmetry is washed out; below $\sim 10^9$ GeV too few triples form out of equilibrium.

The efficiency peaks at $T_{\rm RH} \simeq 1.4 \times 10^{10}$ GeV and keeps its sign throughout. The full window is two orders of magnitude below $\Lambda_{\rm meta}$, consistent with Postulate 3. A bias weighted by triple chirality, $(n_L - n_R)/3$, changes sign between the two binding laws and cannot fix the sign of η_B; the bias must be universal across triples.

Consistency condition. The mechanism reproduces the observed value if

$$\varepsilon\, D \simeq \frac{6.1 \times 10^{-10}}{\eta_B/\varepsilon(T_{\rm RH})}, \tag{4}$$

where $D \leq 1$ is any later entropy dilution. For $D = 1$ this requires $\varepsilon \leq 10^{-5}$ for $T_{\rm RH} \approx 2 \times 10^9$–$2 \times 10^{10}$ GeV, with a minimum of $\varepsilon \simeq 1.7 \times 10^{-7}$ at the peak. Equation (4) is a requirement on the bias, not a prediction. The chirality parameter $\varepsilon_\psi = 25/169$ is CP-even and four to six orders of magnitude larger than required, so it cannot serve as the bias without a suppression that remains to be derived.

5.4 Conditions

The mechanism is identified and its structure verified; its magnitude rests on modelled contact kinetics and on two underived inputs.

(C1) *Rung label.* Derive Postulate 1: a rung acquired at binding that changes only through dissociation. Below $\Lambda_{\rm meta}$ the psions are confined, so a rung change would require passing through the deconfined continuum. A suppression controlled by the condensate action $S = 212.7$ of the PH [7] would satisfy the bound of Corollary 8. Establishing it requires computing the amplitude between rungs of one triple, including the overlap of rungs whose sizes differ by $e^{1/b_e} \simeq 34.6$ per level.

(C2) *The bias.* Show that the bifurcation bias is odd under inversion of the winding orientation, which fixes its sign, and derive its magnitude. The natural carrier is the chiral Jackiw–Rebbi zero mode of the winding condensate [14].

The reheating temperature is an input from inflation, not from the PH. If C2 yields ε, Equation (4) turns the mechanism into a prediction of $T_{\rm RH}$, which is independently constrained by inflationary physics.

6 Predictions

Predictions 1–3 are structural and hold irrespective of the mechanism; 4 and 5 hold if level bifurcation is the origin of the asymmetry.

1. No nucleon decay with $\Delta B = \Delta L$ ($p \to e^+\pi^0$, $p \to e^+\gamma$, ...): these change N_ψ and are forbidden exactly by Theorem 1.
2. If any nucleon decay is observed, it has $\Delta B = -\Delta L$ ($p \to \nu\pi^+$, $n \to \nu\pi^0$, $n \to 3\nu$).
3. No neutrinoless double-beta decay and no n–$\bar{n}$ oscillation: both change N_ψ.
4. A cosmic antineutrino excess of two per baryon, i.e. a neutrino degeneracy parameter $|\xi_\nu| \approx 2.5 \times 10^{-9}$ per flavour (shared over two flavours). This is far below current bounds, and it is the characteristic, if difficult, signature of the mechanism.
5. An inflationary reheating temperature in the range $T_{\rm RH} \approx 10^9$–4×10^{10} GeV, below the condensation scale; outside it the mechanism fails.

7 Discussion

The PH inverts the Standard-Model structure of baryon and lepton number. This places it outside the families of mechanisms that rely on sphaleron reprocessing, and it rules out several routes that would otherwise be natural. The constraint that remains is sharp: the theory needs a conserved baryon label. Without it, the dual-sector identification makes nucleons unstable. With a label acquired at binding, a CP-biased bifurcation yields the observed composition with no tuning of the composition itself. The one parameter beyond the PH, the reheating temperature, has a natural range consistent with the condensation scale.

The conditions fit together. A reheating temperature below Λ_{meta} is what keeps the condensate orientation uniform, and it is also where formation is out of equilibrium with the natural coupling. Confinement below Λ_{meta} is the candidate reason why the rung, once acquired, survives.

Two limitations bound the present results. First, the kinetic networks use contact rates and binding weights that are models, not derived amplitudes. The structural statements (Proposition 10) are independent of these choices; the efficiencies of Table 1 are not. Second, Postulates 1 and 2 are additions to (1) until C1 and C2 are derived.

A related tension concerns dark matter. If the top of the cascade were a thermal relic, $\Omega_{\mathrm{DM}}h^2 = 0.12$ would require a mass of ~ 56 TeV (cascade level ≈ 4.8). The canonical PH value of 7.2×10^{10} GeV would overclose the Universe; a non-thermal production route is therefore required.

8 Conclusions

Equation (1) conserves psion number and CP exactly. Observation then requires separate baryon and lepton numbers with $B + L$ fixed, and the PH must change $B - L$ to produce an asymmetry. Neither the Lagrangian nor its minimal extension does so. Level bifurcation during formation after reheating is a candidate mechanism. If its postulates hold, it gives $B = -L$, charge neutrality and the observed composition by symmetry. With the natural coupling, its efficiency reaches 3.6×10^{-3} per unit bias at a reheating temperature of $\sim 10^{10}$ GeV. Whether it accounts for η_B depends on two derivations: a rung label conserved after binding, and the sign and size of the bias.

Methods and data availability

All tests were defined in advance in documents whose SHA-256 hashes and UTC timestamps were recorded before execution. Controls that failed their registered thresholds are reported as failures in the repository, together with corrections and their reasons. Networks were integrated with LSODA, DOP853 or Radau at relative tolerances of 10^{-10}–10^{-11}. Exact results were obtained by linear algebra over the rationals or by singular-value decomposition. Code, definitions, registries and outputs are available in the companion repository [8], folder `p14d_psion_number_audit`.

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